Google Skills

The solution: Model Context Protocol (MCP)

A universal adapter for tools

What is MCP?

MCP (Model Context Protocol) is an **open standard** created by Anthropic and now hosted by The Linux Foundation. It allows AI agents to connect to external tool servers through a universal protocol. Think of it as USB for AI tools—any MCP-compatible server can provide tools to any MCP-compatible client, regardless of the AI framework being used.

From ADK documentation:

“ADK helps you both use and consume MCP tools in your agents. Refer to the MCP Tools documentation for code samples and design patterns that help you use ADK together with MCP servers.”:

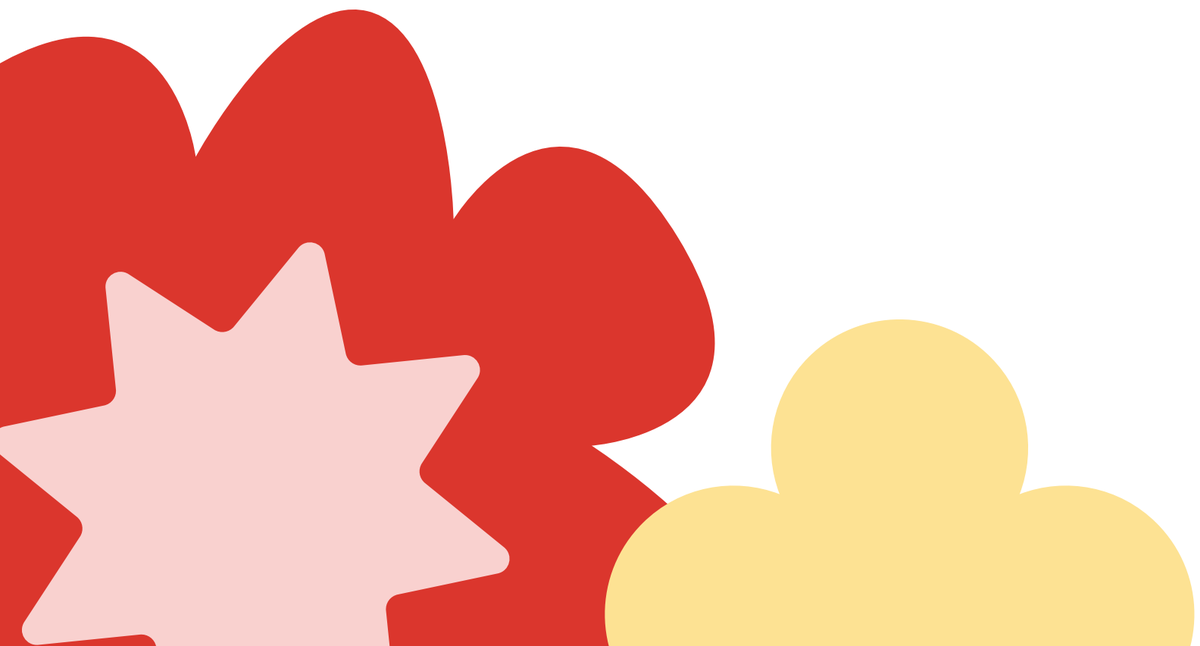
Two integration patterns with ADK:

ADK supports two MCP integration patterns:

1. **Using Existing MCP Servers:** Your ADK agent acts as an MCP client, consuming tools from external servers (this module)
2. **Exposing ADK Tools via MCP:** You build an MCP server that wraps your ADK tools, making them accessible to any MCP client (advanced, see docs)

This module focuses on **Pattern 1**: Connecting to pre-existing MCP servers to extend your agent's capabilities.

Reference: ADK docs - MCP tools | Official MCP documentation



Why MCP matters

The power of standardization:

Without MCP	With MCP
Each tool needs custom integration code	One integration pattern works for all tools
Each tool needs custom integration code	Tools work across Claude, GPT, Gemini, and more
You maintain every integration yourself	Community and vendors maintain their tools
Limited to what's built into your framework	Access to a growing ecosystem of 100s of tools

Industry adoption:

- **Anthropic** created the protocol (November 2024)
- **OpenAI** officially adopted MCP (March 2025)
- **Google ADK** provides native MCP support via `McpToolset`
- **The Linux Foundation** now hosts MCP as an open standard

What this means for you:

- **Write once, use anywhere:** MCP tools work across AI frameworks
- **Leverage the ecosystem:** Use tools built by specialists (databases, APIs, cloud services)
- **Stay current:** As new MCP servers are published, your agent can use them immediately
- **Separation of concerns:** Focus on your agent's logic, let tool maintainers handle integrations

Finding MCP servers

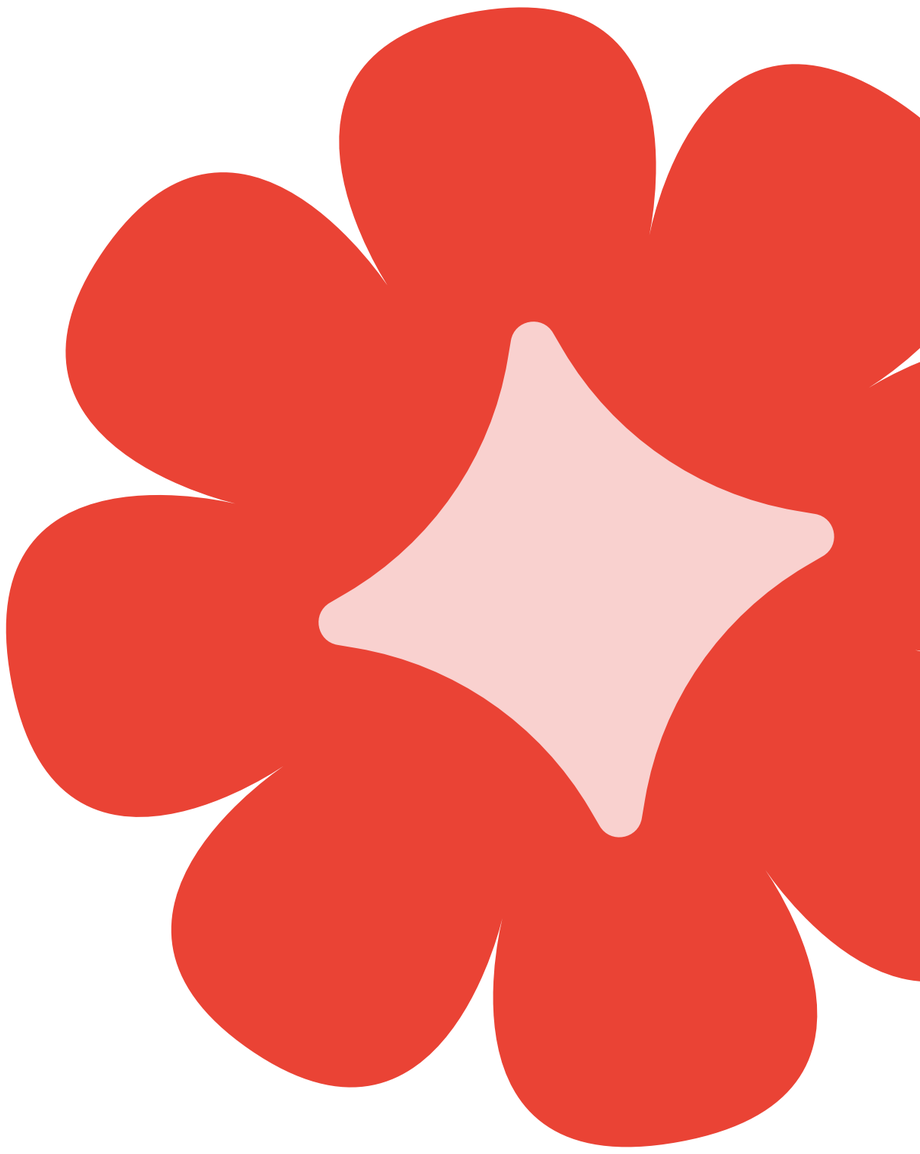
Before building custom tools, check if an MCP server already exists for your use case.

Official resources:

Resource	Description	Link
MCP server registry	Official searchable catalog of MCP servers	registry.modelcontextprotocol.io
Reference servers	Official servers maintained by MCP team	github.com/modelcontextprotocol/servers
MCP documentation	Full protocol specification and guides	modelcontextprotocol.io

Popular MCP servers include:

- **Filesystem:** Read/write files and directories
- **GitHub:** Repository management, issues, PRs
- **Slack:** Send messages, read channels
- **PostgreSQL/MySQL:** Database queries
- **Notion:** Document and database access
- **Google Drive:** File storage access



Tip:

Search the MCP Registry before writing custom tools—the capability you need may already exist.

How MCP works in ADK

Integration flow:

None

```
graph LR
    A[ADK Agent] --> B[McpToolset]
    B --> C[MCP Server]
    C --> D[Tools: list_directory, read_file, etc.]

    style A fill:#e1f5ff
    style B fill:#ffeb99
    style C fill:#eeffee
```

The process:

1. **MCP servers** expose tools via the standardized MCP protocol
2. **ADK agents** connect to servers using `McpToolset`
3. **Tools are discovered automatically**, no manual registration needed
4. **Your agent uses these tools** exactly like any other ADK tool

ADK agents connect to MCP servers through `McpToolset`, automatically discovering available tools

Core concepts

1. The `McpToolset` class

From ADK documentation:

"The `McpToolset` class can be directly added to your agent's tools list, enabling seamless connection to an MCP server, discovery of its tools, and making them available for your agent to use."

Reference: ADK docs - MCP tools

What `McpToolset` does:

1. **Connects** to an MCP server using connection parameters
2. **Discovers** available tools from the server automatically
3. **Proxies** tool calls from your agent to the MCP server
4. **Returns** results back to your agent

How to use:

Python

```python
from google.adk.agents import LlmAgent
from google.adk.tools.mcp_tool import McpToolset
from google.adk.tools.mcp_tool.mcp_session_manager import StdioConnectionParams
from mcp import StdioServerParameters

agent = LlmAgent(
    model='gemini-2.5-flash',
    name='my_agent',
    instruction="Use available tools to help users.",
    tools=[
        McpToolset(
            connection_params=StdioConnectionParams(
                server_params=StdioServerParameters(
                    command='npx',
                    args=['-y', '@some/mcp-server'],
                ),
            ),
        )
    ],
)
```

What to notice:

- `McpToolset` goes in the `tools` list like any other tool
- Connection parameters tell ADK how to reach the MCP server
- Tools from the server become available to your agent automatically

2. Connection types

ADK supports two ways to connect to MCP servers:

`StdioConnectionParams` (local servers)

Launches an MCP server as a local subprocess on your machine.

Python

```python
from google.adk.tools.mcp_tool.mcp_session_manager import StdioConnectionParams
from mcp import StdioServerParameters

# Local server connection
connection = StdioConnectionParams(
    server_params=StdioServerParameters(
        command='npx',                                          # Command to run
        args=['-y', '@modelcontextprotocol/server-filesystem', '/path'],  # Arguments
    ),
)
```

When to use: Development, testing, single-user scenarios.

`SseConnectionParams` (remote servers)

Connects to an MCP server running remotely via HTTP.

Python

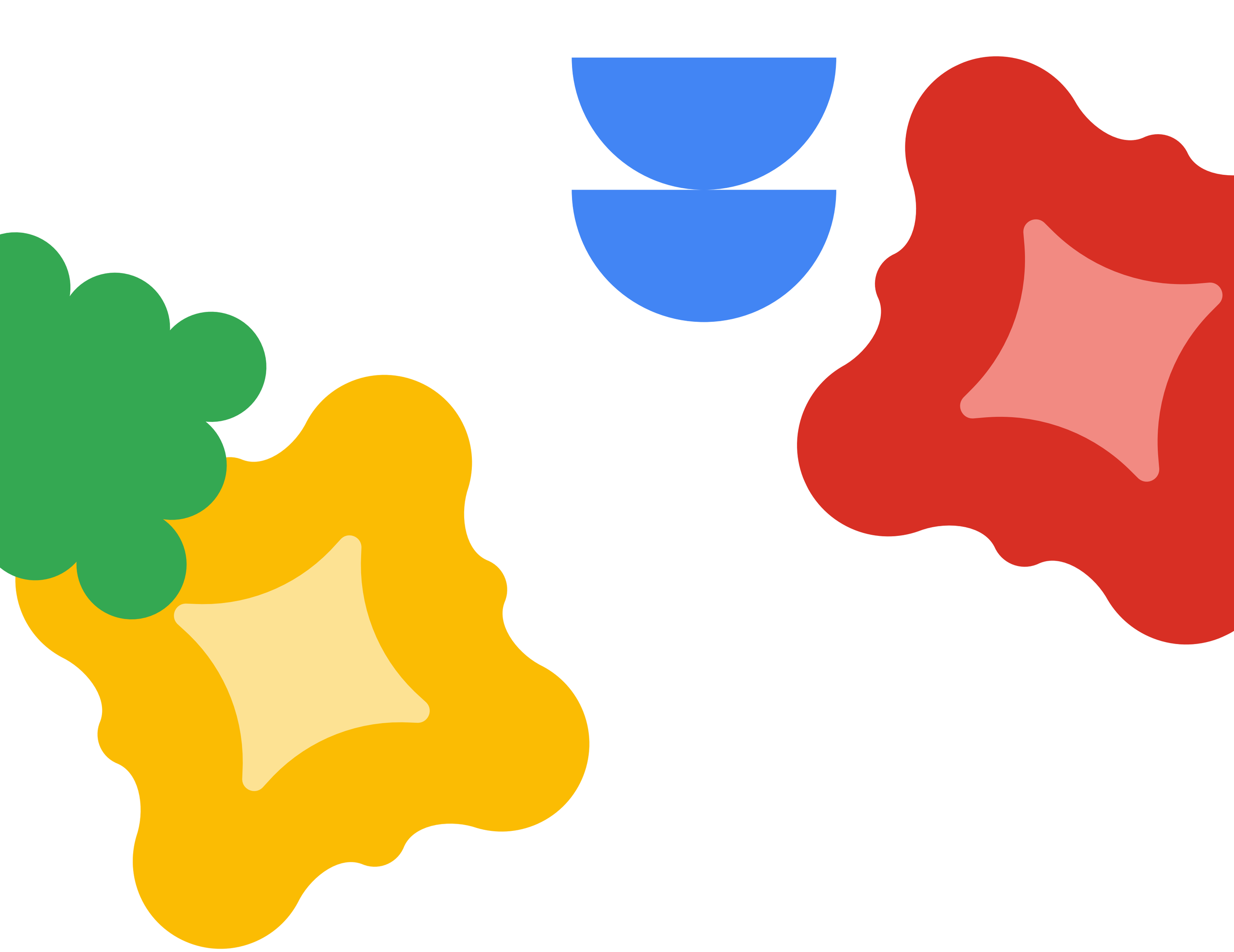
```python
from google.adk.tools.mcp_tool.mcp_session_manager import SseConnectionParams

# Remote server connection
connection = SseConnectionParams(
    url="https://your-mcp-server.example.com/sse",
    headers={'Authorization': 'Bearer YOUR_TOKEN'},
)
```

When to use: Production deployments, cloud-hosted MCP servers.

Reference: ADK docs - MCP tools

Connection types comparison:

Aspect	`StdioConnectionParams`	`SseConnectionParams`
Server location	Local (subprocess)	Remote (HTTP)
Setup	Simple (npx command)	Requires server infrastructure
Best for	Development, testing	Production, cloud
Scalability	Single user	Multiple users



3. Tool filtering

MCP servers may expose many tools, but you might only want your agent to use specific ones. The `tool_filter` parameter lets you control which tools are available.

Industry adoption:

- **Security:** Limit to read-only tools, exclude dangerous operations
- **Simplicity:** Give agent only what it needs for its task
- **Focus:** Reduce confusion from too many tool options

How to use:

```
Python
McpToolset(

connection_params=StdioConnection
Params(...),

tool_filter=['list_directory',
'read_file'],  # Only these tools
)
```

Reference: ADK docs - MCP tools

Example: Restricting filesystem access

```
Python
# ✅ Good: Only expose safe, read-only tools
McpToolset(
    connection_params=StdioConnectionParams(
        server_params=StdioServerParameters(
            command='npx',
            args=['-y', '@modelcontextprotocol/server-filesystem', '/allowed/
path'],
        ),
    ),
    tool_filter=['list_directory', 'read_file'],  # No write access
)

# ❌ Risky: Exposing all tools including write operations
McpToolset(
    connection_params=StdioConnectionParams(
        server_params=StdioServerParameters(
            command='npx',
            args=['-y', '@modelcontextprotocol/server-filesystem', '/'],
        ),
    ),
    # No filter = all tools exposed, including write_file, delete, etc.
)
```

Best practice: Always use `tool_filter` in production to expose only the tools your agent needs.

4. Choosing the right tool type

When should you use MCP tools versus built-in tools or custom function tools?

The decision framework:

```
Do you need this capability?
    |
    ├─→ Is it Google Search or Code Execution?
    |       └─→ YES → Use Built-in Tools (Module 2)
    |
    ├─→ Does an MCP server already exist for this?
    |       └─→ YES → Use MCP Tools (this module)
    |           Check: registry.modelcontextprotocol.io
    |
    └─→ Is this your own business logic or proprietary system?
            └─→ YES → Write Custom Function Tools (Module 4)
```

Key differences that matter:

Aspect	MCP tools	Built-in tools	Custom function tools
Who maintains it	Community/vendors	ADK team	You
Setup effort	Connect and configure	Import	Write code
Portability	Works across AI frameworks	ADK only	ADK only
Customization	Limited to server options	Limited	Full control
Time to implement	Minutes	Minutes	Hours to days

MCP tools are ideal when:

- The capability is **common** (files, databases, popular APIs)
- Someone else has **already solved** the integration problem
- You want **portability** across AI frameworks
- You prefer **maintained tools** over writing your own

Custom function tools are better when:

- The logic is **unique** to your business
- You're connecting to **proprietary** internal systems
- You need **full control** over behavior and error handling
- **No MCP server** exists for your specific use case

Real-world decision example:

Need	Decision	Rationale
Read files from disk	MCP: `@modelcontextprotocol/server-filesystem`	Filesystem access is common; MCP server exists
Query PostgreSQL database	MCP: `@modelcontextprotocol/server-postgres`	Database access is common; MCP server exists
Calculate shipping cost	Custom function tool	Business-specific logic; only you know your rates
Access internal employee API	Custom function tool	Proprietary system; no public MCP server

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Hands-on example

Building a file reader assistant with MCP tools

What you'll build: An agent that can list and read files using the filesystem MCP server.
Prerequisites: Node.js and npm installed (for the npx command)

Step 1: Create the project

```shell
adk create geography_assistant
cd geography_assistant
```

What this does:

- Creates a new ADK project directory
- Sets up the basic agent structure
- Creates `agent.py` with default agent code

Step 2: Create a test folder with files

```shell
mkdir -p my_files
echo "Hello, this is a test file." > my_files/hello.txt
echo "This is another file with different content." > my_files/notes.txt
```

What this does:

- Creates a folder that the MCP server will access
- Adds sample files for the agent to read

Step 3: Write the agent

Replace the contents of `agent.py` with:

```python
"""
File Reader Assistant Agent
Demonstrates MCP tools integration with ADK using the filesystem MCP server.

Reference: https://google.github.io/adk-docs/tools-custom/mcp-tools/
"""

import os
from google.adk.agents import LlmAgent
from google.adk.tools.mcp_tool import McpToolset
from google.adk.tools.mcp_tool.mcp_session_manager import StdioConnectionParams
from mcp import StdioServerParameters

# Define the folder to allow file access (must be absolute path)
ALLOWED_PATH = os.path.abspath("./my_files")

# Create the folder if it doesn't exist
os.makedirs(ALLOWED_PATH, exist_ok=True)

# Create the agent with MCP filesystem tools
root_agent = LlmAgent(
    model='gemini-2.5-flash',
    name='file_reader_assistant',
    description='Helps users read and explore files using MCP tools.',
    instruction="""
    You are a file reader assistant that helps users explore files.
```

```
Your capabilities:
    - List files in directories using list_directory
    - Read file contents using read_file

    When helping users:
    1. Use list_directory to show available files
    2. Use read_file to display file contents when asked
    3. Describe what you find in a helpful way

    Always be clear about which folder you're working with.
    """,
    tools=[
        McpToolset(
            connection_params=StdioConnectionParams(
                server_params=StdioServerParameters(
                    command='npx',
                    args=[
                        '-y',
                        '@modelcontextprotocol/server-filesystem',
                        ALLOWED_PATH,
                    ],
                ),
            ),
            # Filter to only expose safe, read-only tools
            tool_filter=['list_directory', 'read_file'],
        )
    ],
)
```

Code walkthrough

Lines 9–14: Setup and configuration

- Import `McpToolset` and connection classes from ADK
- Import `StdioServerParameters` from the MCP package
- Define `ALLOWED_PATH` as an absolute path to the files folder

Lines 17–18: Create folder

- Ensure the target folder exists before the agent runs

Lines 21–47: Create agent with MCP tools

- `McpToolset` connects to the filesystem MCP server
- `StdioConnectionParams` launches the server via npx
- `tool_filter` limits tools to `list_directory` and `read_file`
- Instruction guides the agent on how to use these tools

Step 4: Run and test

```
Shell
adk web
```

Navigate to `http://localhost:8000` in your browser.

Step 5: Test scenarios

Test 1: List files

You: `What files are in the folder?`

Expected behavior:

- Agent calls `list_directory` tool via MCP
- Tool returns list of files in `my_files` folder
- Agent presents the file list to you

What to notice:

- Agent discovered and used the MCP tool automatically
- No custom tool code was needed—the MCP server provided everything

Test 2: Read a file

You: `Show me the contents of hello.txt`

Expected behavior:

- Agent calls `read_file` tool with the file path
- Tool returns the file contents
- Agent presents the contents to you

What to notice:

- Agent correctly invoked the MCP tool with the right argument
- File contents are returned through the MCP protocol

Test 3: Multiple operations

You: `List the files and then read the notes.txt file`

Expected behavior:

- Agent calls `list_directory` first
- Agent then calls `read_file` for notes.txt
- Agent presents both results together

What to notice:

- Agent orchestrates multiple MCP tool calls
- Same patterns you learned in Module 1 apply to MCP tools

Key takeaways

What MCP is and why it matters:

- **Open standard** for connecting AI agents to external tool servers
- **Universal interoperability:** Works across AI frameworks (ADK, Claude, GPT)
- **Industry adoption:** Anthropic, OpenAI, Google, and The Linux Foundation
- **Ecosystem access:** Hundreds of pre-built tools for common tasks

Using MCP in ADK:

- `McpToolset` connects your agent to MCP servers
- **Tools are discovered automatically** - no manual registration
- **Use like any other tool** - same patterns from Module 1 apply

Connection types:

- `StdioConnectionParams` - Local servers via subprocess (development)
- `SseConnectionParams` - Remote servers via HTTP (production)

Security with tool filtering:

- **Use** `tool_filter` to expose only needed tools
- **Limit to read-only** when possible for safety
- **Fewer tools = clearer decisions** for your agent

When to use each tool type:

Scenario	Approach	Why
Pre-existing capability (files, databases, APIs)	MCP tools	Community maintains it
Google capabilities (search, code execution)	Built-in tools	ADK team maintains it
Your business-specific logic	Custom function tools	Only you know your business

Key resources:

- **MCP Server Registry:** Find existing MCP servers
- **Official MCP docs:** Protocol documentation
- **ADK MCP guide:** ADK integration docs

Beyond this module: Advanced MCP capabilities

This module covered the fundamentals of using existing MCP servers. MCP and ADK offer additional capabilities for more advanced use cases.

What else you can do with MCP in ADK:

Capability	Description	Documentation
Create your own MCP server	Wrap your custom tools in an MCP server to share with others	ADK docs - Building MCP servers
Expose ADK tools via MCP	Make your ADK tools accessible to any MCP client (Claude, GPT, etc.)	ADK docs - MCP tools
Connect to multiple MCP servers	Use tools from several MCP servers in a single agent	ADK docs - MCP tools
Deploy MCP servers to Cloud Run	Host your MCP servers for production use	ADK docs - MCP tools

The second integration pattern (advanced):

Earlier, we mentioned ADK supports two MCP integration patterns.
The second pattern—**exposing ADK tools via an MCP server**—allows you to:

1. Wrap your existing ADK tools in an MCP server
2. Make those tools available to any MCP-compatible client
3. Share your tools with the broader AI ecosystem

This is valuable when you want other AI frameworks (Claude, GPT, etc.) to use tools you've built in ADK.